

**HINDUSTAN PETROLEUM
CORPORATION LIMITED**



**TENDER DOCUMENT
FOR
PIPELINE INSPECTION AT COIMBATORE IRD**

INTRODUCTION

Hindustan Petroleum Corporation Limited (HPCL) proposes to engage a reputed third-party inspection agency for carrying out the External on-stream Inspection of Pipelines at HPCL Coimbatore IRD

HPCL installations have large number of pipelines within the operating depot/ terminals. Inspection of these Pipelines at regular interval is essential in ensuring good health of these critical assets. Pipelines are laid between storage tanks to loading gantries through pump house connections. We also operate pipelines from our installation to BPCL Irugur Terminal for product receipt. The length of the pipelines is around 350 m and these pipelines cross the railway mainline in underground section. Suitable methods to be employed for the inspection of these pipelines are well. Similarly, the fire hydrant network is also laid around the depot/ terminal.

HPCL wishes to carry out the inspection of these Pipelines at Coimbatore IRD as per OISD 130, API 570, API 579 & other statutory requirements as applicable. Pipelines are

- a. Above ground Product Pipelines of various sizes
- b. Above ground Fire Water/Foam Pipelines of various sizes

As these Pipelines are in service for a long period of time, HPCL wishes to check fitness of these pipelines for further service thru various inspection methods by qualified and experienced inspection personnel. Based on the recommendations given by inspecting agency, HPCL will carry out the necessary repairs. Inspection agency shall also carryout post repair inspection of the tank to give fitness certificate.

NOTE: The bidders are advised to visit site to familiarise themselves with actual site conditions and scope of work before quoting.

BID QUALIFICATION CRITERIA:

The bidders should have carried out similar works (Pipeline Inspection confirming to OISD-130 & relevant API-570, API-579 latest editions for pipelines) in Oil & Gas installations in India.

The bidder should have completed at least 1 similar work of minimum 13.18 Lakhs in the Last 7 years ending on the date 31.03.2025. The bidder to submit relevant completion certificate/POs in this regard.

ROLE OF INSPECTION AGENCY

The following are the responsibilities of the inspection agency:

- To inspect, measure and record the deterioration/corrosion of materials and to evaluate current Physical conditions of the pipelines for soundness and its continuation in service.
- To keep the concerned operating and maintenance personnel fully informed as to the condition of the various pipelines.
- To co-relate the deterioration/corrosion rate with design life for further run of pipelines.
- To investigate the causes of deterioration and recommend remedial measures such as short term and long-term repairs/ replacements as per OISD-130 & relevant API-570, API-579 latest editions for pipelines.
- To suitably advise the inspection methods, and to carry out the inspection of the terminal-to-terminal pipelines of around 350 m from HPCL Coimbatore IRD to BPCL Irugur Terminal which crosses the Railway mainline in underground section.

Details of Pipelines:

The pipelines are of various sizes up to maximum diameter of 24 inches. The tentative lengths of all pipes clubbed together is around 25000 meters.

Inspection of these network of pipelines at regular interval is essential in ensuring good health of these critical assets. HPCL wishes to carry out the inspection of following types of pipelines at Coimbatore IRD in line with OISD-130, API 570, API 579 & other statutory requirements as applicable.

- a. Above ground Product Pipelines of various sizes

- b. Above ground Fire Water Pipelines of various sizes
- c. Foam line

As these pipelines are in service for around 3 years, HPCL wishes to check fitness of these pipelines for further service thru various inspection methods by qualified and experienced inspection personnel.

TOOLS REQUIRED FOR INSPECTION

Tools required in general for tank inspection are as follows:

- 1) Ultrasonic Thickness Meter
- 2) Radiographic Equipment
- 3) LRUT Equipment
- 4) DCVG Equipment
- 5) Dye Penetrant Kit
- 6) Holiday Detector
- 7) Shore Hardness Meter
- 8) Paint Thickness Gauge
- 9) Vacuum Box Tester
- 10) Safety Torch
- 11) Knife / Scraper
- 12) Crayon
- 13) Magnifying Glass
- 14) Permanent Magnet
- 15) Vernier Calliper / Micrometer
- 16) Fillet Gauge
- 17) Measuring Tape / Scale
- 18) Straight Edge
- 19) Pit Gauge
- 20) Plump Bob and Line

SITE INSPECTION

Site inspection in general will involve following inspection methods:

- 1) Visual Inspection (VI)
- 2) Close Visual inspection (CVI)
- 3) Dye-Penetrant Testing (PT)
- 4) Ultrasonic Testing (UT)

- 5) External On-stream Inspection of Above Ground (AG) pipelines
- 6) Inspection of Fire Water sprinklers and Foam risers.
- 7) LRUT for inaccessible Pipelines.
- 8) DCVG for inaccessible UG pipelines with wrapping coating.
- 9) Advanced Ultrasonic Testing i.e. TOFD / Phased Array, if required.

INSPECTION PROCEDURES

Before commencing the inspection of a Pipelines, all details given in its history card and records shall be gone thorough.

EXTERNAL ON-STREAM INSPECTION

Visual Inspection

Leaks

Visual inspection shall be made to locate leaks. Particular attention should be given to pipe connections, the packing glands of valves and expansion joints.

Alignment

The piping shall be inspected for misalignment. Inspector shall make note of following indications in the inspection report for each pipeline section being inspected.

- a) Pipe dislodged from its support so that the weight of the pipe is distributed unevenly on the hangers or the saddles.
- b) Deformation of the wall of the vessel in the vicinity of the pipe attachment.
- c) Pipe supports forced out of plumb by expansion or contraction of the piping.
- d) Shifting of base plate or shearing of the foundation bolts of mechanical equipment to which the piping is attached.
- e) Cracks in the connecting flanges or pump casings to which the piping is attached.

Supports

Pipe supports shall be visually inspected for the following:

- a) Condition of protective coating or fire proofing if any. If fire proofing is found defective, sufficient fire proofing should be removed to determine extent of corrosion.
- b) Evidence of corrosion
- c) Distortion
- d) General physical damage
- e) Movement or deterioration of concrete footings.

- f) Condition of foundation bolts.
- g) Free operation of pipe rollers.
- h) Secure attachment of brackets and beams to the supports.
- i) Secure attachment and proper adjustment of pipe hangers, if used. Spring hangers loading shall be checked both cold and hot and the readings obtained shall be checked against the original cold and hot readings. The movement of spring supports shall be monitored.
- j) Broken or otherwise defective pipe anchors.
- k) Free operation of pulleys or pivot points of counter balanced piping system.

Vibrations

- a) If vibrations or swaying is observed, inspection, shall be made for cracks in welds, particularly at points of restraint such as where piping is attached to equipment and in the vicinity of anchors. Additional supports should be considered for poorly braced small size piping and valves and for main vibrating line to which they are attached.
- b) In case of severe vibration, detailed investigations shall be carried out to determine the source of problems.

External Corrosion

Inspection of piping for external corrosion shall be carried out.

Bulging, Bowing and Sagging

Line shall be checked for bulging, bowing and sagging in between the supports.

Mechanical Damage from External Forces

Pipes shall be inspected for dents, scratched etc. from external sources.

Paint and Protective Coating

Conditions of paint and protective coating shall be checked.

Cracks

Pipelines shall be inspected for cracks. Particular attention should be given to areas near the weld joints.

Insulation

Damage of insulation shall be checked.

Concrete Lining

Externally concrete lined piping shall be visually inspected for cracking and dislodging of concrete.

Ultrasonic Thickness Survey

Ultrasonic thickness survey of the pipelines shall be carried out to ascertain the wall thickness. The following procedure shall be followed for the aboveground pipelines.

- a) Minimum three readings shall be taken on the bends of the piping at the outer curvature. One reading shall be at the centre of the bend and two readings on the same line on either side of this reading.
- b) Minimum one ultrasonic scan each on the straight pipes on upstream and downstream of the bend adjacent to welds of the bend to pipe. One ultrasonic scan consists of four readings (3, 6, 9 and 12 o'clock positions).
- c) For Fire Water Pipelines & other product lines in which there is a possibility of ballast water coming, one ultrasonic scan will consist of six readings (3, 5, 6, 7, 9 and 12 o'clock positions) to scan the bottom portions.
- d) One ultrasonic scan on the entire circumference (four readings) upstream and downstream of the weld joint.
- e) Minimum one ultrasonic scan (four readings) each on reducer/ expander and just downstream on the pipe.
- f) One ultrasonic scan on the pipe downstream of valves orifices, etc.
- g) One ultrasonic scan minimum on horizontal pipe for every three meters length at lower elevations where possibilities of collection and stagnation of carryover water exist.
- h) Branch connection, dead ends etc, shall be checked by ultrasonic thickness survey for corrosion and erosion.

- i) The details of thickness survey shall be maintained on an isometric sketch.
- j) Measurement of thickness reduction on the corrosion coupon, if any, shall be done to establish the corrosion rate. Past data of UT readings will be shared by HPCL.

THICKNESS MEASUREMENT OF INACCESSIBLE AREAS USING THE LRUT TECHNIQUE.

Long range ultrasonic guided wave technology (LRUT) should be deployed to assess metal loss in inaccessible location not covered under conventional ultrasonic thickness measurement. Following areas are considered as inaccessible for ultrasonic thickness testing.

- a) All Pipelines passing through the culverts
- b) All underground sections of the pipelines

Technique shall be based on Torsional Ultrasonic waves or longitudinal waves and it shall be capable of checking condition of the Pipelines. Party shall submit the credentials related to successful application of technique / procedure for similar work from reputed users like Hydrocarbon Processing Industry or Refinery. Party shall submit a brief approach Paper indicating details of technique & instrumentation to be employed by them.

Instrument shall be battery operated. Agency shall arrange for sufficient cable so that the computer can be stationed at convenient place while carrying out the tests at various locations.

Agency shall submit a final comprehensive report containing observations for each pipe inspected showing detailed location of areas of low thickness. Report shall be in hard copy form as well in soft form in a CD/pendrive.

Agency shall have its own provision of calibration of equipment / instrument. The agency shall demonstrate the calibration before commencement of job.

Subsequent to the identification of Metal loss, UT Gauging / scanning of the area to be carried out by the agency for verification and LRUT data mapping of the internal/external corrosion and to assess the exact remaining thickness of the pipe, the area shall be exposed by HPCL in case-to-case basis.

SCANNING OF SPRINKLER/ FOAM RISERS & RINGS

In each location, product storage tanks are provided with sprinklers & foam pourers for firefighting purpose. Due to corrosive medium (Water & Foam), these lines are likely to have high internal corrosion rates. In view of criticality of such lines, 100% scanning of such lines shall be done using LRUT technique.

These lines sizes may vary from upto 3 to 6" inch depending on the size of the tank.

- a) Bidder to scan the 100% Vertical Risers of sprinkler / foam line of each tank.
- b) Sprinkler & foam rings within the dyke shall be scanned 100% using LRUT technique.
- c) HPCL shall provide working platform wherever required.
- d) Bidder to ensure that all the instruments except the line clamping device shall be placed at ground level away from the tank.

Hammer Testing

Hammer testing shall also be carried out to supplement visual and ultrasonic inspection. While hammer testing, the following precautions shall be taken:

- a) Only non-sparking material hammer shall be used.
- b) Hammer testing of valves, pipes and fittings of cast iron construction, chrome-steel, austenitic SS lines and stress relieved lines shall not be carried out.
- c) Care shall be taken not to hammer so hard as to damage otherwise sound piping.
- d) Hammer testing shall not be performed on glass-lined, cement-lined or otherwise internally coated lines.
- e) Hammer testing shall not be done on operating lines and lines under pressure.

Coating Survey of the U/G pipeline sections

Most of the underground pipeline sections are coated with bitumen-based coating materials to protect these pipelines against external deterioration caused by corrosive soil condition. Condition of the coating shall be checked with the help of Direct Voltage

Gradient survey (DCVG) method. There may be 20 to 30 numbers of underground sections of pipelines in terminal. Each underground section may be in the range of 5 to 10m for each grade of product/Water.

- a) Conducting Direct Current Voltage Gradient (DCVG) Survey in one meter interval and as per direction of EIC.
- b) Also locating the exact position of defect for physical verification.
- c) Party has to submit the detailed procedures for carrying out DCVG Survey and line locating work etc. Party should submit the list of manpower employed with them along with experience, qualification, name, age etc. The work shall commence only after the Engineer in-Charge approves the procedures submitted by the party.
- d) Party should give details of equipment proposed to be used for this survey along with the name of the equipment, model, and year of manufacturing and data card for each equipment. Party should clearly indicate the availability of DCVG instruments along with Make, Model, and Year etc.
- e) For the purpose of DCVG survey, party shall carryout soil resistivity survey at pipe depth at suitable distance from the pipeline in the depot/terminal. No additional cost is permissible for above surveys.
- f) Work shall be carried out as per standard practice and International Standards and specifications.
- g) To ascertain exact location, size and magnitude of coating defect throughout the periphery of the pipeline by carrying out the DCVG survey at the probable defect locations. Contractor has to quantify and categorize the coating defects in the terms of severity as Low, moderate & severe.

REPORT

Detailed report shall be made for each Tank and Pipeline section inspected showing all the UT readings taken at all positions.

Isometrics/ piping plan of each circuit as per actual site conditions shall be prepared. The records shall be maintained to give information like:

- a) Identification of particular piping systems in terms of location, class, total length, material specification, general process flow and service condition and location of corrosion probes, if any.
- b) Location of thickness measurement points, replacement carried out, corrosion rate, etc.
- c) All small connections shall be clearly shown on isometrics and piping drawings.

All testing and inspection reports with UT readings shall be submitted in one soft copy & two hard copies duly signed by the inspecting engineer.

COMPLETION SCHEDULE

The completion schedule for **Pipeline inspection** job is **Two months** from the date of LOA/PO whichever is earlier which will also include submission of 'Site Assessment Report'.

Any delay beyond the above completion schedule, Corporation shall be entitled to recover from the bidder, liquidated damages @ 0.5% of charges per week of delay subject to a maximum of 5% of the Charges for the delay, if the reasons are solely attributable to the bidder.

Completion schedule is the essence of contract, and the bidder will make all out efforts to ensure completion within the time schedule.

Pipelines Inspection at each location will involve following stages:

- Kick-off meeting with HPCL site engineer/location in-charge before start of work at site
- Collection and review of site data and site inspection to determine the actual condition of assets
- Identification of the entire damage mechanism operative in each equipment and selection of the relevant NDT methods based on operative damage mechanism
- Selection of relevant inspection type as per OISD-130/ API-570, API-579
- Preparation of Activity schedule and execution
- Review of data/drawings/ Material certificates/ operating history & previous NDT inspection reports provided by HPCL including information about wall thickness measurements, damages or defects discovered etc.
- Review of Repair reports and NDT reports after repairs, if any

- Review of WPS (Welding procedure specification) and Welding procedure qualification testing (WPQT) reports of all relevant original and repair welds, if available
- Any other relevant information

PAYMENT TERMS: Payment shall be released site wise only after completion of the full Assessment work of each site & submission of the final report of that site.

Deliverables

- a) Draft site report with findings and recommendations to HPCL for review and comments within a week

Submission of Final report to HPCL both in soft and hard copies and will include the following:

- 1) Pipeline segment wise detailed assessment results
- 2) Checklist as per OISD-130, API 570 and 579 as applicable.
- 3) Decision to run/repair/alter or replacement of the equipment
- 4) Guidance of the inspection interval of the Tanks and Pipeline sections based on the remaining life
 - i. Determination of the feasibility of continuation in service due to increased severity of the operation
 - ii. Remaining life evaluation of Tanks & Pipelines based on established corrosion rate.
 - iii. Recommendation of in-service monitoring and techniques required for maintaining the tanks and the facility

Security Deposit:

2% of PO/Contract value as Security deposit will be acceptable in the form of Demand draft or through e-payment upto Rs 50,000/- and in the form of Demand draft / Bank guarantee or through e-payment beyond Rs 50,000/-.

Composite Performance Bank Guarantee (CPBG) for 5% of PO value towards Performance Bank Guarantee inclusive of Security Deposit shall be accepted (in lieu of deduction of retention money); such CPBG shall be valid up to a period of 3 months beyond the expiry of defect liability period.

RETENTION MONEY

5% of the total value of the Running Account and Final Bill will be deducted and retained by the Owner as retention money on account of any damage/defect liability that may arise for the period covered under the defect liability period clause of the Contract free of interest.

The retention money is not applicable, if the Bidder provides Performance Bank Guarantee (PBG) for 5% of the Job / contract value.

HPCL's SCOPE

HPCL shall provide the following:

- a. All the available data of the Tanks & Pipelines under inspection shall be provided by the Location EIC, once successful bidder visits the location.
- b. All the tanks shall be thoroughly cleaned and degassed by HPCL prior to handing over to contractor for Inspection.
- c. Site clearance around the tank and pipeline areas for thorough inspection shall be done by HPCL.
- d. HPCL will carry out the excavation work, wherever required for bell hole inspection.
- e. Removal of coating, buffing of the weld joints and other severely corroded areas as identified jointly during the inspection will be done by HPCL. Recoating after successful completion of verification and Ultrasonic testing on such areas will be done by HPCL.
- f. Provision of adequate light, ventilation inside the tank and fan shall be provided by HPCL.

CONTRACTOR'S SCOPE

Apart from HPCL's obligation as mentioned above, all other obligations and responsibilities are with the contractor to complete the entire job as per terms and conditions of the tender document. This will include but not limited to the following:

Deployment of adequate manpower, skilled and unskilled

All manpower (experts, technicians, non-technical, skilled, unskilled, labour) required for the above job shall be in the scope of bidder and they shall deploy sufficient manpower at site.

Team of expert shall consist of

- 1) Project Manager
- 2) Deployment of NDT experts having level-3 certification
- 3) Deployment of NDT technicians with level-2 certification, along with equipment and their valid calibration certificates.

Tools & Tackles

All the tools and tackles, equipment, consumables, required to complete the above job shall be in the scope of the bidder and their mobilization and demobilization after completion of works shall be in the scope of the bidder.

Lodging/Boarding/Transportation

Lodging, boarding and transportation of bidder's representatives shall be in the scope of the bidder.

SPECIAL TERMS & CONDITIONS

- a. The offer should be valid for 90 days from the due date of this tender.
- b. Scope of the work has been covered in various sections/Appendices enclosed to this tender document. Subletting of the whole or any part of the work is not allowed.
- c. This is a Time Bound Project & any delay in providing the Comprehensive Assessment report will adversely affect the Tank outage plan. In case it is observed that the vendor fails to give the services within the specified time and acceptable manner, HPCL

reserves the right to terminate the Appointment with immediate effect and complete the balance works through an alternate agency at the entire risk & cost of the vendor.

- d. The effective date of the contract will be the date of Letter of Acceptance/PO whichever is earlier.
- e. SAFETY COMPLIANCE AT SITE/LOCATION: Inspection agency and its personnel shall fully comply with all safety related requirements at location. Inspection personnel shall wear helmets, Safety shoes and required PPEs while carrying out inspection of storage tank. Instructions given by Location incharge/Site engineer in this regard shall be fully adhered to.

AWARDING OF CONTRACT:

The contract shall be awarded to a single party with lowest quote on overall basis.

Schedule of Quantities

Quantity mentioned in SOQ is probable quantity and thus HPCL is not confirming utilisation of full quantity as mentioned in the SOQ

Sr No	Item	QTY	UOM
1	TESTING OF WALL THICKNESS OF ABOVE GROUND PIPELINE: ULTRASONIC THICKNESS SURVEY OF THE PIPELINES OF ALL DIA UPTO 24" SHALL BE CARRIED OUT TO ASCERTAIN THE WALL THICKNESS. THE FOLLOWING PROCEDURE SHALL BE FOLLOWED FOR THE ABOVE-GROUND PIPELINES. A) MINIMUM THREE READINGS SHALL BE TAKEN ON THE BENDS OF THE PIPING AT THE OUTER CURVATURE. ONE READING SHALL BE AT THE CENTER OF THE BEND AND TWO READINGS ON THE SAME LINE ON EITHER SIDE OF THIS READING. B) MINIMUM ONE ULTRASONIC SCAN EACH ON THE STRAIGHT PIPES ON UPSTREAM AND DOWNSTREAM OF THE BEND ADJACENT TO WELDS OF THE BEND TO PIPE. ONE ULTRASONIC SCAN CONSISTS OF FOUR READINGS (3, 6, 9 AND 12 O'CLOCK POSITIONS). C) FOR FIRE WATER PIPELINES AND OTHER PRODUCT LINES IN WHICH THERE IS A POSSIBILITY OF BALLAST WATER COMING, ONE ULTRASONIC SCAN WILL CONSIST OF SIX READINGS (3, 5, 6, 7, 9 AND 12 O'CLOCK POSITIONS) TO SCAN THE BOTTOM PORTIONS. D) ONE ULTRASONIC SCAN ON THE ENTIRE CIRCUMFERENCE (FOUR READINGS) UPSTREAM AND DOWNSTREAM OF THE WELD JOINT.	21000	Running Meter

	E) MINIMUM ONE ULTRASONIC SCAN (FOUR READINGS) EACH ON REDUCER/ EXPANDED AND JUST DOWNSTREAM ON THE PIPE. F) ONE ULTRASONIC SCAN ON THE PIPE DOWNSTREAM OF VALVES ORIFICES, ETC. G) ONE ULTRASONIC SCAN MINIMUM ON HORIZONTAL PIPE FOR EVERY THREE METERS LENGTH AT LOWER ELEVATIONS WHERE POSSIBILITIES OF COLLECTION AND STAGNATION OF CARRYOVER WATER EXIST. H) BRANCH CONNECTION, DEAD ENDS ETC, SHALL BE CHECKED BY ULTRASONIC THICKNESS SURVEY FOR CORROSION AND EROSION. I) THE DETAILS OF THICKNESS SURVEY SHALL BE MAINTAINED ON AN ISOMETRIC SKETCH. J) MEASUREMENT OF THICKNESS REDUCTION ON THE CORROSION COUPON IF ANY, SHALL BE DONE TO ESTABLISH THE CORROSION RATE. PAST DATA OF UT READINGS WILL BE SHARED BY HPCL.		
2	TESTING OF METAL LOSS AND TESTING OF COATING DAMAGES OF UNDERGROUND/INACCESSIBLE PIPELINE LONG RANGE ULTRASONIC GUIDED WAVE TECHNOLOGY (LRUT) SHOULD BE DEPLOYED TO ASSESS METAL LOSS IN INACCESSIBLE LOCATION NOT COVERED UNDER CONVENTIONAL ULTRASONIC THICKNESS MEASUREMENT.FOLLOWING AREAS ARE CONSIDERED AS INACCESSIBLE FOR ULTRASONIC THICKNESS TESTING. A) ALL PIPELINES PASSING THROUGH THE CULVERTS. B) ALL UNDERGROUND SECTIONS OF THE PIPELINES. 1. TECHNIQUE SHALL BE BASED ON TORSIONAL ULTRASONIC WAVES OR LONGITUDINAL WAVES AND IT SHALL BE CAPABLE OF CHECKING CONDITION OF THE PIPELINES. PARTY SHALL SUBMIT THE CREDENTIALS RELATED TO SUCCESSFUL APPLICATION OF TECHNIQUE / PROCEDURE FOR SIMILAR WORK FROM REPUTED USERS LIKE HYDROCARBON PROCESSING INDUSTRY OR REFINERY. PARTY SHALL SUBMIT A BRIEF APPROACH PAPER INDICATING DETAILS OF TECHNIQUE AND INSTRUMENTATION TO BE EMPLOYED BY THEM. 2. INSTRUMENT SHALL BE BATTERY OPERATED. AGENCY SHALL ARRANGE FOR SUFFICIENT CABLE SO THAT THE COMPUTER CAN BE STATIONED AT CONVENIENT PLACE WHILE CARRYING OUT THE TESTS AT VARIOUS LOCATIONS. 3. AGENCY SHALL SUBMIT A FINAL COMPREHENSIVE REPORT CONTAINING OBSERVATIONS FOR EACH PIPE INSPECTED SHOWING	4000	Meters

DETAILED LOCATION OF AREAS OF LOW THICKNESS. REPORT SHALL BE IN HARD COPY FORM AS WELL IN SOFT FORM IN A CD.

4. AGENCY SHALL HAVE ITS OWN PROVISION OF CALIBRATION OF EQUIPMENT / INSTRUMENT. THE AGENCY SHALL DEMONSTRATE THE CALIBRATION BEFORE COMMENCEMENT OF JOB.
5. SUBSEQUENT TO THE IDENTIFICATION OF METAL LOSS, UT GAUGING / SCANNING OF THE AREA TO BE CARRIED OUT BY THE AGENCY FOR VERIFICATION AND LRUT DATA MAPPING OF THE INTERNAL/EXTERNAL CORROSION AND TO ASSESS THE EXACT REMAINING THICKNESS OF THE PIPE, THE AREA SHALL BE EXPOSED BY HPCL IN CASE-TO-CASE BASIS.

THE CONDITION OF THE COATING SHALL BE CHECKED WITH THE HELP OF DIRECT VOLTAGE GRADIENT SURVEY (DCVG) METHOD. THERE MAY BE 5 TO 10 NUMBERS OF UNDERGROUND SECTIONS OF PIPELINES IN THE TERMINAL. EACH UNDERGROUND SECTION MAY BE IN THE RANGE OF 5 TO 10M FOR EACH GRADE OF PRODUCT.

- a) CONDUCTING DIRECT CURRENT VOLTAGE GRADIENT (DCVG) SURVEY IN ONE METER INTERVAL AND AS PER DIRECTION OF EIC.
- b) ALSO LOCATING THE EXACT POSITION OF DEFECT FOR PHYSICAL VERIFICATION.
- c) PARTY HAS TO SUBMIT THE DETAILED PROCEDURES FOR CARRYING OUT DCVG SURVEY AND LINE LOCATING WORK ETC. PARTY SHOULD SUBMIT THE LIST OF MANPOWER EMPLOYED WITH THEM ALONG WITH EXPERIENCE, QUALIFICATION, NAME, AGE ETC. THE WORK SHALL COMMENCE ONLY AFTER THE ENGINEER IN-CHARGE APPROVES THE PROCEDURES SUBMITTED BY THE PARTY.
- d) PARTY SHOULD GIVE DETAILS OF EQUIPMENT PROPOSED TO BE USED FOR THIS SURVEY ALONG WITH THE NAME OF THE EQUIPMENT, MODEL, AND YEAR OF MANUFACTURING AND DATA CARD FOR EACH EQUIPMENT. PARTY SHOULD CLEARLY INDICATE THE AVAILABILITY OF DCVG INSTRUMENTS ALONG WITH MAKE, MODEL, AND YEAR ETC.
- e) FOR THE PURPOSE OF DCVG SURVEY, PARTY SHALL CARRYOUT SOIL RESISTIVITY SURVEY AT PIPE DEPTH AT SUITABLE DISTANCE FROM THE PIPELINE IN THE DEPOT/TERMINAL. NO ADDITIONAL

	COST IS PERMISSIBLE FOR ABOVE SURVEYS. SOIL RESISTIVITY SURVEY MAY BE SLIGHTLY AWAY FROM THEM PIPELINE. f) WORK SHALL BE CARRIED OUT AS PER STANDARD PRACTICE AND INTERNATIONAL STANDARDS AND SPECIFICATIONS. g) TO ASCERTAIN EXACT LOCATION, SIZE AND MAGNITUDE OF COATING DEFECT THROUGHOUT THE PERIPHERY OF THE PIPELINE BY CARRYING OUT THE DCVG SURVEY AT THE PROBABLE DEFECT LOCATIONS. CONTRACTOR HAS TO QUANTIFY AND CATEGORIZE THE COATING DEFECTS IN THE TERMS OF SEVERITY AS LOW, MODERATE AND SEVERE.		
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ADDRESSES OF TERMINAL

Terminals

Facility Name : Coimbatore IRD
Location : Coimbatore
Site : Tamil Nadu
Address : HPCL Coimbatore IRD
AG Pudur, Irugur
Coimbatore - 641103

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